

CSE 260 LAB REPORT 1

Experiment name:

Familiarisation of Fundamental
Logic Gates

Group no. 5

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Section: 22

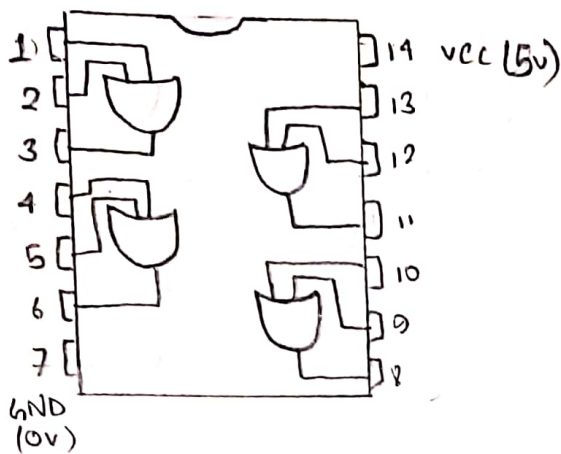
OBJECTIVE :

- To get familiarised with fundamental logic gates and demonstrate the input output relationship of 2-input AND (IC-7408), OR (IC-7432) and NOT (IC-7404) gates by constructing their truth tables.
- To get familiarised with fundamental logic gates like NAND (IC-7400), NOR (IC-7402), XOR (IC-7486) and XNOR (IC-74266)

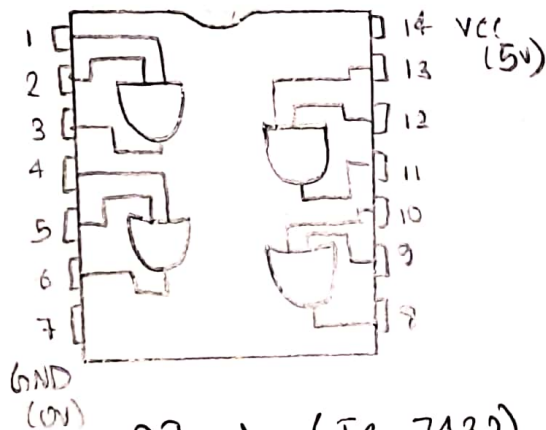
Required components and equipments :

- Seven IC
[IC-7408, IC-7432, IC-7404, IC-7400, IC-7402, IC-7486, IC-74266]
- Wires
- Breadboard
- Portable digital/analog laboratory
From the portable lab we need:
 - power supply
 - ground supply
 - switches
 - LED lights

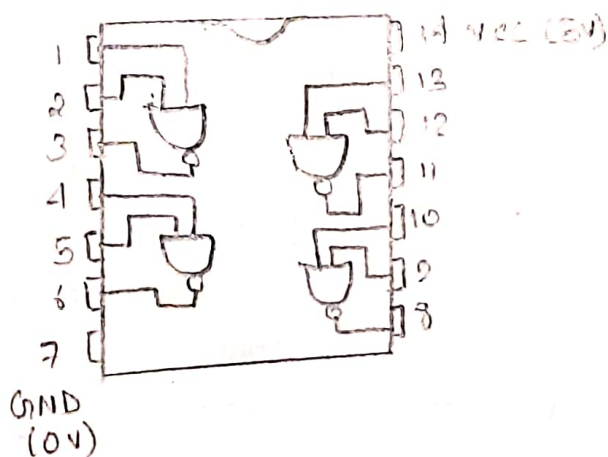
Experimental Setup:



AND gate (IC-7408)



OR gate (IC-7432)

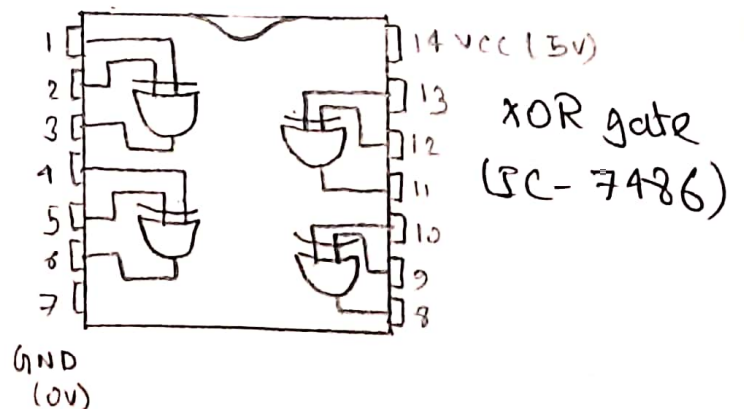


NAND gate
(IC-7400)

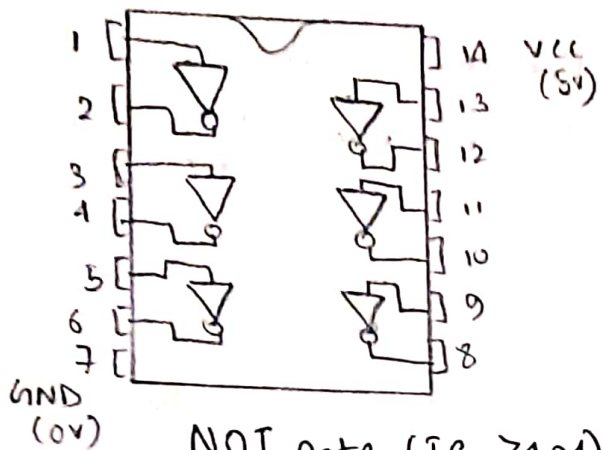
For the logic gates OR, XOR, NAND and AND gate, the IC is connected vertically on the ~~the~~ e-f column of the breadboard. Pin 7 is connected to the negative column, which is ~~the~~ then ~~com~~ connected to the ground supply (GND) of the portable digital lab.

+5V from the digital lab is connected to the positive column, where we also connect pin 14.

Pin 1 and 2 are taken as input and are connected to the switches 1 and 2 respectively on the digital lab. Pin 3 is the output pin and it is connected to a LED light port using suitable wires.

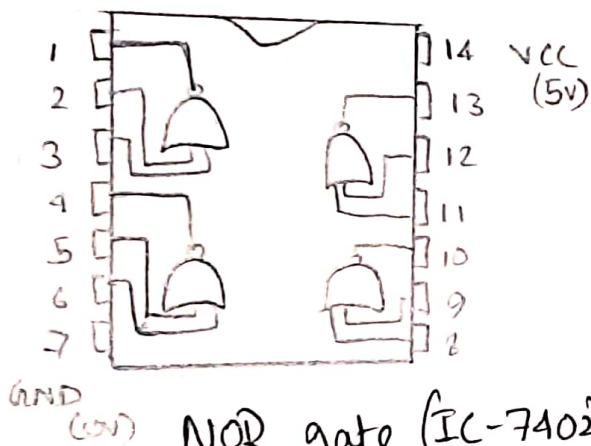


XOR gate
(IC-7486)



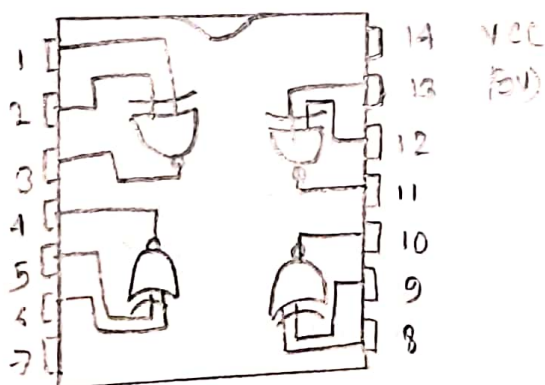
NOT gate (IC-7404)

For NOT gate, pin 7 again is connected to GND and pin 14 to 5V (VCC) in the same way as before. Pin 1 here is the input and is connected to switch 1 and pin 2 is connected to LED light of the portable digital lab.



NOR gate (IC-7402)

For NOR gate, pin 7 is connected to GND and pin 14 to 5V again. Here pin 2 and 3 are input and are connected to switch 1 and 2 and pin 1 is the output that is connected to the LED.



XNOR gate (IC-74266)

For XNOR gate, a combination of XOR gate and NOT gate is used. A XOR gate is setup as shown above. The output from XOR gate is then given as input for the NOT gate. Lastly the output of NOT gate is connected to the LED.

TRUTH TABLES

AND gate (IC-7408)

Input		Output
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

NOT gate (IC-7404)

Input	Output
A	
0	1
1	0

NOR gate (IC-7402)

Input		Output
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

Alvin
12.10.22

Alvin
12.10.22

IC-
OR gate (7432)

Input		Output
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

NAND gate (IC-7400)

Input		Output
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

XNOR gate IC-74266

Input		Output
A	B	
0	0	1
0	1	0
1	0	0
1	1	1

XOR gate (IC-7486)

Input		Output
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

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Alhina
12.10.22

Discussion

For the AND gate, the LED light only glows when both of the switches is turned on.

For the OR gate, the LED light glows at all combinations except when both switches are turned off.

For the NOT gate, there is only a single switch. The bulb glows when the switch is off only.

For the NAND gate, the bulb glows for all combinations except when both the switches is turned on. It is basically the opposite of AND gate.

For XOR gate, the LED bulb glows when one of the switches is turned on. It does not glow when both are on or both are off.

For NOR gate, the light only glows when both the switches are turned on. It is the opposite of OR gate.

For XNOR gate, the bulb only glows when both the switches are either off or on. It is the opposite of XOR gate.